

Quantify uncertainty in behavioral research

The behavioral sciences underestimate the uncertainty of research findings and thus overestimate replicability. Metrologists in the physical sciences quantify all material components of uncertainty, even if some components must be quantified using non-statistical means. Behavioral science should follow suit.

Edward E. Rigdon, Marko Sarstedt and Jan-Michael Becker

Suppose that researchers were free to ignore some of the variability in their research outcomes. Their reported standard errors would be too small, falsely characterizing results as being ‘statistically significant’. More unquantified variability would increase the chances of extreme results, which scholarly journals like to publish¹, though the larger variability would limit the ability to successfully replicate results.

That is behavioral science today. Researchers report standard errors for their measurements—the quantified values obtained from research—but not the uncertainty of their measurands, which are the actual quantities that they are attempting to obtain. If researchers were required to quantify uncertainty, then existing incentives would no longer perpetuate practices that generate high uncertainty in measurands and poor replication success. Quantifying uncertainty would drive researchers to improve the design of new studies, reducing uncertainty over time and improving our ability to anticipate the replicability of research.

In learning to quantify uncertainty, behavioral science can follow the example of metrology—‘measurement science’ in the physical sciences, engineering and legal forensics². Decades ago, metrologists in several of these disciplines moved the concept of uncertainty to the center of their research practice. Today, in these fields, no measurement is complete without a quantification of the overall uncertainty associated with the underlying measurand. In the following, we argue that behavioral science should follow the same route.

What is uncertainty?

In metrology, uncertainty is “the dispersion of the values that could reasonably be attributed to the measurand.”³ Suppose a researcher wants to quantify the effect of a treatment on attitude toward object X. One hundred randomly sampled individuals are randomly assigned to treatment or control groups, and their attitudes are assessed using a single questionnaire item, which is scored

from 1 to 7. The measurement here is the difference in mean scores for the two groups in the experiment. The measurand is the difference in attitude across the population for those exposed to the treatment versus those not exposed.

The value of the measurand is unknowable. A sound measurement provides a best estimate, but that estimate always leaves uncertainty regarding the value of the measurand. High uncertainty means that the measurement is consistent with a wide range of plausible values for the measurand, and researchers may expect to observe a wide range of values across different measurements and across different studies. This does not mean that the measurand has multiple values, in the sense of a random coefficient model, but only that the researcher’s limited knowledge leaves a range of values as plausible.

Uncertainty arises from many sources across the entire research process, including from incomplete definition of the measurand, from an unstable measurand and from limitations of the measurement instrument. In behavioral science, measurements rarely involve laboratory tools, and uncertainty comes mainly from the human participants—both those providing data and those collecting data. Random sampling error, as expressed in statistical standard errors, is one important component of overall uncertainty³, but it is not necessarily the largest. Importantly, uncertainty cannot be less than the standard error alone, unless researchers have ignored some phenomenon like finite population in computing their standard error.

Metrology recognizes two general approaches to quantifying components of uncertainty. Type A is the familiar statistical approach, making multiple observations under specified conditions and estimating uncertainty based on variation across observations. When a Type A method cannot be used, for example because the necessary studies are economically or ethically infeasible, metrologists turn to Type B methods—using prior data, reference values, individual experience or scientific judgment³.

An uncertainty budget collects the available information on all material components of uncertainty⁴. When those components are valued, overall uncertainty is calculated in the same way that s.d. are combined—square each s.d. to make a variance, sum the variances, then take the square root (and account for any covariances between components). Researchers then multiply overall uncertainty by a coverage factor, chosen to reflect a desired level of confidence, to create a coverage interval³. Choosing a coverage factor to achieve an interval with precise probability can be complex, involving *t* distributions with approximated degrees of freedom², but it often suffices to assume an approximately normal distribution, so that a coverage factor of 2 yields a confidence of approximately 95% (ref. ³).

Managing uncertainty in behavioral science

Of course, behavioral science already quantifies uncertainty, just haphazardly. Statistical standard errors from a single study capture variability across respondents, which may be due to variation in the substantive phenomena and variation in the execution of research procedures within the study. But these statistical estimates do not capture variation that is likely to occur across research teams⁵, as when independent researchers attempt replication.

Checklists, such as those now mandatory for *Nature* journals, represent one approach for managing components of uncertainty. Calls to expand the ‘methods’ section of research reports, to enable potential replicators to more closely reenact research procedures, are another. Requiring researchers to assume smaller Type I error rates for “new discoveries”⁶ recognizes that original research in novel settings may be more subject to variation in procedures and thus higher uncertainty. But such steps still leave it to the reader to construct an understanding of the full uncertainty of findings. Limiting uncertainty by imposing constraints or by raising thresholds is important, but only quantifying and

Table 1 | Example uncertainty budget

Uncertainty component	Degrees of freedom	Type	Probability distribution	Divisor	Standard uncertainty	Relative contribution
Random sampling	98	A (s.d. of the mean)	Normal	1	0.6	61.79%
Resolution (rating scale)	Inf.	B (half width between scale points)	Uniform	$\sqrt{3}$	0.289	14.30%
Measurement error	Inf.	B (20% error variance of total variance)	Normal	1	0.072	0.89%
Omission of part of the population	Inf.	B (18% of population missing)	Normal	1	0.210	7.57%
Definitional uncertainty	Inf.	B (expert assessment)	Normal	1	0.3	15.45%
Combined uncertainty	$\sqrt{0.6^2 + 0.289^2 + 0.072^2 + 0.21^2 + 0.3^2} = 0.763$					

Note: Following the *Guide to the Expression of Uncertainty of Measurement*¹, a convention is to assign infinite (inf.) degrees of freedom to s.d. derived from Type B methods.

managing uncertainty proactively will improve research practice over time.

So, what will we need to quantify uncertainty across behavioral science? We will need standardized definitions of the phenomena that we study, to help limit definitional uncertainty.

We need standard operating procedures. For metrologists, even simple length measurements are associated with an extensive body of guidance intended to limit uncertainty. By contrast, behavioral scientists aiming to capture unobservable mental phenomena invent their procedures again and again.

We need standard uncertainty budgets, to help researchers to recognize and account for components of overall uncertainty. Reconsider the study introduced above, which may be subject to substantial uncertainty beyond sampling error as identified by the following uncertainty budget (Table 1). Suppose that, for example, attitude is a continuous variable that prior research has shown to be normally distributed in the population of interest, and suppose that the study measures a mean difference in attitude of 0.3, with s.d. = 0.6. That would yield a *t* value of 2.5 or a 95% confidence interval of [0.062; 0.538]—a ‘significant’ result in a frequentist analysis.

But there are other sources of uncertainty. Psychometricians are concerned about the presence of ‘measurement error’. Perhaps prior applications of the single indicator suggest a reliability of 80%, meaning that 20% of all variance is error variance. Uncertainty can be estimated through the connection between uncertainty and factor indeterminacy⁷. If the total variance of the observed variable is 0.36 (= 0.6²), then the uncertainty due to measurement error is $0.20 \times 0.36 = 0.072$.

Another potential source of uncertainty comes from non-coverage of the population.

Assume that the sampling approach fails to represent 18% of the intended population. With no information about attitude difference in the excluded part of the population, it could lie anywhere on the 7-point scale. As we assume that attitude is normally distributed, the s.d. can be approximated by dividing the 7-point range by 6 (ref. ²). The uncertainty is then estimated as $18\% \times (7 \div 6) = 0.21$.

The limited precision of the scale also contributes uncertainty. Without any further information, the measurand could lie variously anywhere in the interval from one scale point to the next, so that the uniform distribution provides a convenient approximation. The standard uncertainty is then estimated as one-half the interval divided by the square root of 3, or 0.289³. Other uncertainty components, such as the definitional uncertainty resulting from the finite amount of detail in the definition of a measurand, are more difficult to quantify and require scientific judgment.

Only taking this—certainly not exhaustive—list of uncertainty components into account, the initial 95% confidence interval becomes [−0.003; 0.603], thereby increasing 27% in width. In a frequentist analysis this result would not be ‘statistically significant’ anymore. Of course, adding more uncertainty components related to, for example, the measurand (for example, social desirability bias), the operator (interviewer bias), the method (distributional assumptions) and the environment (noise or temperature variations in the laboratory) will further widen the interval.

Subjectivity is inherent in any quantification of Type B uncertainties, so some elements of our uncertainty budget are certainly debatable. To identify the manifold sources of uncertainty and increase objectivity in their quantification (for example, regarding distributional

assumptions), we need research. Properly designed replication studies like the open science Many Labs projects, in which multiple teams apply the same protocol to study the same phenomenon⁸, could provide important data for establishing standard values for common uncertainty components. In some physical science disciplines, there are research centers whose primary purpose is to quantify uncertainty. We need centers like that in behavioral science.

Above all, we need a requirement and expectation that researchers quantify and report uncertainty. Without a mandate, inertia and the real costs of quantifying uncertainty will undermine progress.

Potential objections

There are certainly substantial obstacles to overcome in giving the quantification of uncertainty a central role in behavioral science practice. In learning to quantify uncertainty, behavioral science cannot simply adopt the metrological procedures of the physical sciences. Those procedures have been created and refined for a world where measurands are ‘well-defined’³ and where measurements are taken using laboratory equipment. When researchers want to measure aspects of human perception (like the gloss on a painted surface), which can only be indirectly linked to laboratory equipment readings, they use modified procedures which fall under the heading of ‘soft metrology’. To extend this notion to the psychological phenomena of behavioral science, researchers will need to create a complete new set of procedures.

Some researchers may be put off by the need to rely on uncertainty quantifications based on Type B evidence, as these may be viewed as suspiciously subjective³. But the real practice of science involves a hundred subjective decisions—pure objectivity is a mischaracterization⁹. We know that

our results are uncertain, but instead of acknowledging the sources of uncertainty, we arbitrarily assign a value of zero to any component not captured by our standard errors. We need to apply the same care in quantifying uncertainty as we apply to every other aspect of our practice.

Properly quantifying and reporting the uncertainty of our behavioral science findings is not optional—we cannot continue pretending that our findings are more certain than we know they are. Yes, creating an infrastructure for assessing uncertainty calls for a substantial investment of time, money and expertise. However, going forward with our research without the ability to quantify all material components

of uncertainty is an unacceptable betrayal of our scientific mission. □

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Competing interests

The authors declare no competing interests.

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